Distributed Edge Artificial Intelligence

A Thesis (Major Project) submitted in partial fulfilment of the requirements

For the Degree of

Bachelor of Technology

In

Computer Science and Engineering

Submitted by

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Under the Guidance of

Mr. Rana Singha

[Supervisor designation]



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
BARAK VALLEY ENGINEERING COLLEGE
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[Month–Year]

Declaration

We, the students of Computer Science & Engineering, Barak Valley Engineering College, declare that the work entitled "Distributed Edge Artificial Intelligence" has been completed under the guidance of Mr. Rana Singha, [Supervisor designation], Computer Science & Engineering, Barak Valley Engineering College.

The report has been prepared without resorting to plagiarism. We have adhered to the principles of academic honesty and integrity. No falsified or fabricated data have been presented in this report. Further, the matter embodied in this project report has not been submitted previously for the award of any degree or diploma.

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Date: _____ Place: _____



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(A Govt. of Assam Institution, Approved by AICTE, New Delhi, Govt. of India)

District: Sribhumi, Assam, 788701

CERTIFICATE OF APPROVAL

Project Title: Distributed Edge Artificial Intelligence
Project Category: Major Project Report
Student Names:

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Name of the Department: Computer Science & Engineering
Name of the Supervisor: Mr. Rana Singha
Academic Session: [Academic session]
Recommendation: [Yes/No]

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CERTIFICATE FROM EXTERNAL EXAMINER

This is to certify that I have examined the major project report entitled "Distributed Edge Artificial Intelligence", submitted by the following students of Bachelor of Technology in Computer Science and Engineering:

1. Anirbar Saikia
2. Avirup Roy
3. Hisbul Bahar Mazumdar
4. Ritesh Das
5. Syed Yashin Hussain

The report is submitted in partial fulfilment of the academic requirements for the degree.

External Examiner: _____

Signature: _____ Date: _____



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CERTIFICATE FROM SUPERVISOR

This is to certify that the major project/thesis work embodied in this report entitled "Distributed Edge Artificial Intelligence", submitted by:

1. Anirbar Saikia
2. Avirup Roy
3. Hisbul Bahar Mazumdar
4. Ritesh Das
5. Syed Yashin Hussain

of Bachelor of Technology in the Department of Computer Science & Engineering is based on their work under my supervision and is prepared for their academic requirements. It is also certified that this work has not been submitted elsewhere for any degree or diploma.

Date: _____ **Mr. Rana Singha**

Place: _____ [Supervisor designation]

Abstract

Formatting sample only — replace before submission. This sample abstract demonstrates the report's paragraph spacing and text width. Distributed Edge Artificial Intelligence studies how computation and learning tasks can be placed across devices near data sources and more centralized computing resources. The final abstract should briefly state the project problem, objectives, approach, evaluation method, and conclusions.

Replace this sample with a concise abstract based on the completed project; do not treat this text as a project result.

Acknowledgement

Formatting sample only — replace before submission. The authors would like to thank their supervisor, Mr. Rana Singha, for guidance during the project. We also acknowledge the support of the Department of Computer Science & Engineering and everyone who contributed advice or feedback.

Update this acknowledgement to reflect the actual contributions and permissions before submission.

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Chapter 1

Introduction

Formatting sample only — replace before submission. Artificial intelligence applications increasingly need to process information close to where it is produced. Distributed Edge Artificial Intelligence considers how data processing and model execution can be shared between resource-constrained edge devices and other computing nodes. This chapter introduces the topic and provides sample section styles for the report.

1.1 Background

An edge-based system may include sensors, user devices, gateways, and computing infrastructure. The placement of a task can affect response time, network traffic, privacy, and resource use. These considerations motivate studying the design choices involved in distributing AI workloads.

1.2 Problem Statement

The project investigates how an AI workload can be organized across edge resources while accounting for device limitations and application requirements. The final problem statement should identify the specific application, constraints, and gap addressed by the project.

1.3 Objectives

The following are sample objective statements; revise them to match the agreed project scope:

1. Study the key concepts and architectures used in distributed edge AI.
2. Define a suitable system design for the selected application.
3. Evaluate the design using project-specific metrics and experiments.

1.4 Scope and Report Organization

This sample document uses five chapters to demonstrate the report structure. The final scope, chapter names, and organization should follow the actual project and the department's requirements.

Chapter 2

Literature Survey

Formatting sample only — replace before submission. This chapter should review research and technical material directly relevant to the selected project. A well-structured survey groups sources by topic, compares their approaches, and explains how the review informs the project design. Add verified citations and a bibliography once the sources have been selected; no reference entries are fabricated in this formatting sample.

2.1 Edge Computing and AI

Introduce the concepts required to understand the project. Define important terms consistently and distinguish background information from the contributions made by the project.

2.2 Comparison of Existing Approaches

Use a consistent basis for comparing related work, such as deployment setting, processing location, evaluation method, and stated limitations. Support each comparison with an appropriate source.

When several approaches are discussed, organize them around common themes rather than presenting a sequence of unrelated summaries. A comparison table can help readers see which assumptions and evaluation settings differ across the reviewed work.

2.3 Research Gap

Conclude the survey by identifying the specific question that remains relevant to this project. The gap should be supported by the reviewed sources and should lead directly to the objectives and design choices described in the following chapters.

Chapter 3

System Design and Methodology

Formatting sample only — replace before submission. This chapter demonstrates how a figure, numbered sections, and a short procedural list can be presented. The diagram below is a generic illustrative workflow and does not describe an implemented system.

3.1 Conceptual Workflow

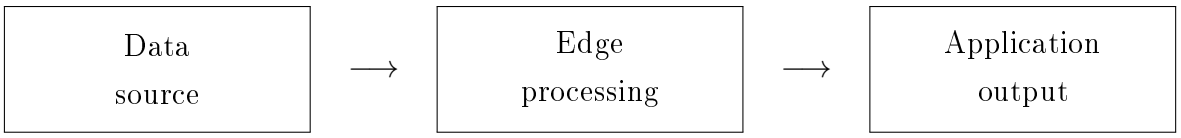


Figure 3.1: Illustrative workflow used only to demonstrate figure formatting.

Figure 3.1 is a placeholder example. Replace it with a diagram that accurately represents the proposed or implemented system.

3.2 Sample Method

An example methodology outline is:

1. Define the application requirements and system constraints.
2. Select the devices, models, and workload distribution strategy.
3. Implement the design and document the experimental setup.
4. Evaluate the system using clearly defined metrics.

3.3 Evaluation Plan

The evaluation plan should describe the test environment, input workload, baseline, and measurement procedure. Define each metric before reporting results so that readers can interpret the measurements consistently.

Chapter 4

Results and Discussion

Formatting sample only — replace before submission. Present the actual observations from the completed experiments in this chapter. State the test conditions, report units, and explain the limitations of the evaluation. The entries in Table 4.1 are intentionally marked as placeholders and are not measured results.

4.1 Illustrative Results Table

Table 4.1: Placeholder results table for demonstrating layout only.

Evaluation case	Metric 1	Metric 2
Case A	TBD	TBD
Case B	TBD	TBD

4.2 Discussion

Replace the placeholder table with measured data and discuss the observed trends in relation to the project objectives. Avoid drawing conclusions that are not supported by the evaluation.

4.3 Limitations

Document constraints that may affect interpretation, such as the size of the test workload, the devices available, or the duration of the experiments. Explain how these constraints were handled and avoid generalizing beyond the conditions that were tested.

Chapter 5

Conclusion and Future Work

Formatting sample only — replace before submission. The final chapter should summarize what was completed and relate the main conclusions to the stated objectives. It should distinguish demonstrated outcomes from proposed future work. This sample contains no project-specific findings.

Keep the conclusion focused: briefly restate the problem, summarize the approach, and state only findings established by the report. Avoid introducing new evidence or claims that have not been discussed in the results.

5.1 Future Work

Possible future-work topics should follow from the actual results and limitations of the project. Replace this paragraph with relevant next steps after the evaluation is complete.

References

[Add the verified references used in the report. Apply the citation style required by the department.]